H.P. DEEGHAU VIDARSHANA PREMASIRI

DATA SCIENCE UNDERGRADUATE

4th Year (4Y1S) IT Undergraduate | Specializing in Data Science
Sri Lanka Institute of Information Technology (SLIIT), Malabe

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I'm a Data Science undergraduate at the Sri Lanka Institute of Information Technology (SLIIT) and currently working as a Data Analyst Intern at Michelin Lanka. I have hands-on experience in data analysis, machine learning, business intelligence, and process automation, using tools such as Python, SQL, Power BI, and Scikit-learn. At Michelin, I've worked on operational analytics, dashboard development, and automation projects, while also helping introduce Dataiku to the plant and guiding employees in its use. Through my academic and professional projects, I've gained experience in predictive modeling, ETL, and data warehousing. I enjoy solving real-world problems, exploring new technologies, and turning complex data into practical solutions that make a difference.

PROFESSIONAL EXPERIENCE

Data Analyst Intern | Michelin Lanka - Ekala Tyre Division 01
June 2026 – Dec 2026 (Present) | Sri Lanka

- Develop Power BI dashboards for manufacturing and operational reporting, supporting performance monitoring and business decision-making.
- Provide Power BI training and guidance to employees, strengthening dashboard adoption and data literacy.
- Help lead the initial adoption of Dataiku at plant level, guiding employees in enterprise analytics workflows.
- Use Dataiku to analyze manufacturing defect patterns and energy consumption behavior.
- Design and develop Microsoft Power Automate flows for local business projects, supporting data processing and reporting automation.
- Develop Power Automate workflows involving international operational data, supporting cross-country data processing and analysis.

TECHNICAL SKILLS

- **Business Intelligence & Visualization:** Power BI, Microsoft Excel, Tableau, Dashboard Development, Data Visualization, Operational Reporting
- **Programming & Data Analytics:** Python, SQL, R, Pandas, NumPy, Exploratory Data Analysis (EDA), Data Cleaning, Data Transformation
- **Data Engineering & Warehousing:** SQL Server, SSIS, SSAS, ETL Pipelines, Data Warehousing, Dimensional Modeling
- **Automation & Enterprise Analytics:** Microsoft Power Automate, Dataiku, Databricks, Workflow Automation
- **Machine Learning & Statistical Analysis:** Scikit-learn, Regression, Classification, Random Forest, Gradient Boosting, Predictive Modeling, Time-Series Analysis, Cross-Validation, Hyperparameter Tuning, Model Evaluation
- **Development Tools:** Git, GitHub, Jupyter Notebook, VS Code, SQL Server Management Studio, REST APIs

PROJECTS

Retail Data Warehouse & Business Intelligence Solution
***2026* | SQL Server, SSIS, SSAS, Power BI, Excel**

- Designed and developed a retail data warehouse using SQL Server, integrating sales data into structured fact and dimension tables for business intelligence reporting.
- Implemented ETL pipelines using SSIS for data extraction, cleansing, transformation, and loading, applying dimensional modeling and Slowly Changing Dimension (SCD Type 2) techniques.
- Developed multidimensional OLAP analysis using SSAS and interactive Power BI dashboards to visualize sales performance, business trends, and key performance indicators.

BankGuard AI — Multi-Agent Intelligent Banking System
***2026* | Python, React, REST APIs, Multi-Agent AI, NLP, RAG**

- Developed an AI-powered banking assistant using a multi-agent architecture to process customer inquiries and provide intelligent, context-aware banking assistance.
- Integrated specialized AI agents for intent recognition, information retrieval, banking product recommendations, eligibility assessment, and response generation.
- Developed a React.js frontend integrated with FastAPI backend services and MongoDB, incorporating security considerations and explainable AI-driven responses.

Hotel Booking Cancellation Prediction — Machine Learning System
***2026* | Python, Pandas, NumPy, Scikit-learn, Random Forest, Gradient Boosting, Matplotlib**

- Developed a machine learning classification workflow using approximately 119,390 hotel booking records, performing exploratory data analysis, missing-value handling, categorical encoding, and leakage-aware preprocessing.
- Trained and evaluated classification models using stratified 5-fold cross-validation, ROC-AUC evaluation, and hyperparameter optimization through RandomizedSearchCV.
- Achieved 0.8945 ROC-AUC, 76.58% precision, and 54.65% recall using the tuned Random Forest model on the held-out test dataset.

StockFlux — Stock Market Prediction & Financial Analytics System
***2025* | Python, Pandas, Scikit-learn, Random Forest, Gradient Boosting, Financial APIs**

- Developed a machine learning-based stock market forecasting system using historical financial datasets, exploratory data analysis, and time-series feature engineering.
- Implemented Random Forest and Gradient Boosting algorithms to analyze market patterns and generate stock price predictions, evaluating model performance using RMSE.
- Integrated financial market APIs to retrieve updated stock information and support dynamic forecasting dashboards and financial data visualization.

EDUCATION

Sri Lanka Institute of Information Technology (SLIIT) - Malabe Oct 2023 – Present

- BSc (Hons) in Information Technology - Specializing in Data Science

Aquinas College of Higher Studies - Colombo 10 Mar 2023 - May 2025

- Diploma in English Language and Literature (Completed)

De Mazenod College - Kandana Jun 2020 – Sep 2023

- G.C.E Advanced Level - Physical science stream

LEADERSHIP & ADDITIONAL SKILLS

- **Technical Leadership:** Helped lead the initial adoption of Dataiku at Michelin Lanka, providing guidance to employees.
- **Project Management:** Led a five-member software development team, coordinating project integration, collaboration, and demonstrations.
- **Knowledge Sharing:** Experience providing Power BI training and communicating technical concepts to colleagues.
- **Languages:** English and Sinhala.

VOLUNTEER EXPERIENCE

- Announcer & Event Coordinator — Faculty of Computing Media Unit (FCMU), SLIIT.
- Member of STEMUP Foundation, Sri Lanka.

REFERENCES

Mr. Tharaka Indunil
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